Business Intelligence – Lab 3

19-20/05/2026 – EPF – A. CHARLEMAGNE

Abstract:

Nota Bene: This document is designed as an autonomous capsule. It contains lecture parts – explaining concepts – and lab parts – practical applications through coding exercises.

⚠️ **Important remark**: This capsule will use a set of icons and graphical chart to indicate several levels of information.

- 👉: Something that you should **do** to complete the Lab
- 💾: You need to **validate** something (call the teacher, save something for deposit)
- 💡: Tips or **useful** remark to help you in your lab
- 🧠: An explanation on a new knowledge, that you should probably know/**retain**
- ⚠️: Something important that you **need** to read
- ⭐: Something to try if you want to **explore** further on the subject
- IT SHOULD NOT BE NEEDED TO BE WRITTEN BUT EVERYTHING ON WHITE BACKGROUND SHOULD BE READ ALSO! (Only ⭐ is optional).

Preamble: Project environment

Welcome to your **Business Intelligence capstone project**!
Throughout this module, you won't just *learn* BI — you will **build your own BI solution**, just like in a real Data team.
You will work **in pairs** and progress step by step during the lab sessions.

🎯 The mini-project concept

During the project, you will define a business need (your choice), select a dataset, and then build:

- A requirements document
- An agile backlog
- A complete BI pipeline (ingestion → transformation → model)
- A final dashboard

By the end, you will be able to lead a BI project from A to Z, just like in a company.

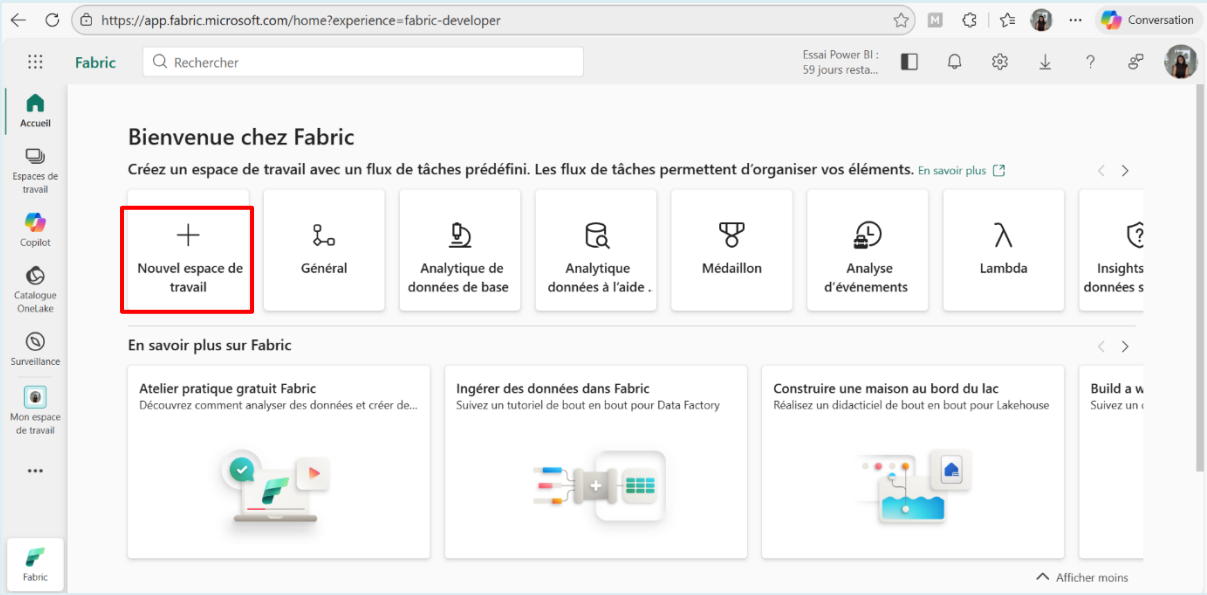
✨ Lab 3 – Data processing

👉 Take the **Word document** from the **last lab session**; we will enrich it today!

Task 1: Create your Microsoft Fabric Workspace

👉 Go to: https://app.fabric.microsoft.com/

👉 Create a new **Workspace** named ***Name1_Name2_BI***



⚠️ You **must activate the free trial (60 days)**.

👉 Open **Manage access** and add:

- Your teammate

👉 Set him as **Administrator**

⚠️ **Important**
All development for this project will be done on Microsoft Fabric.
Make sure your account is correctly set up with the trial activated.

Task 2: Identify required Fabric Elements

⚠️ **Important**
In this lab, you are setting up **only the development environment (DEV)**.

In real BI projects, we usually have:
- DEV (development)
- TEST / UAT
- PROD (production)

👉: You need to **design your BI architecture**. From the list below, choose the components you need:

- Data Pipeline
- Notebook
- Dataflow Gen2
- Lakehouse
- Warehouse

👉 For each selected component, explain:

✅ What is its role

✅ Why you chose it in your architecture

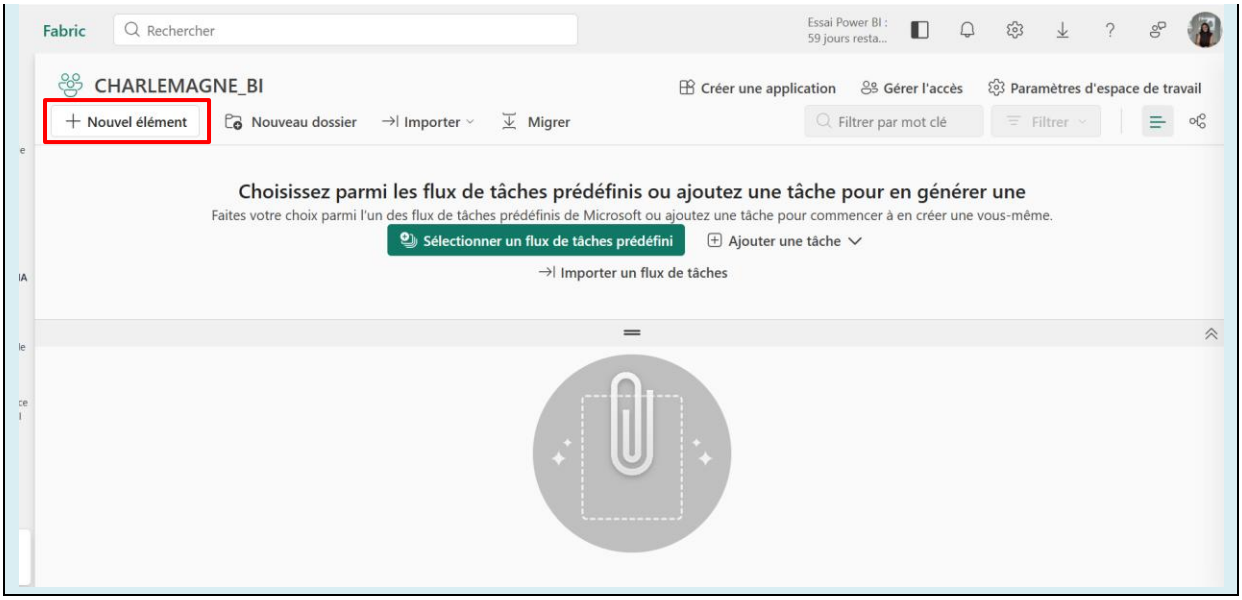
✅ Why you did NOT choose the others

💡 Example:

We use a Lakehouse to store raw and transformed data in a scalable way.

💡 No need to use all components. Think **simple but coherent architecture.**

👉 Then, create the selected elements in your workspace (using **“New item / Nouvel élément”**)



💡 Remember about nomenclature

Lakehouse: LH_XX

Datawarehouse: DWH_XX

Dataflow: DF_XX

Pipeline: PL_XX

Notebook: NB_XX

Task 3: Implement Medallion Architecture

You will now structure your data using the **Medallion Architecture**:

- 🥉 Bronze → raw data
- 🥈 Silver → cleaned data
- 🥇 Gold → business-ready data

💡 Example

🥉 Bronze:

Sales_raw (original dataset with all columns)

🥈 Silver:

Sales_clean (fixed types, no missing values, standardized categories)

🥇 Gold:

Fact_Sales (aggregated sales data)

Dim_Product (product information)

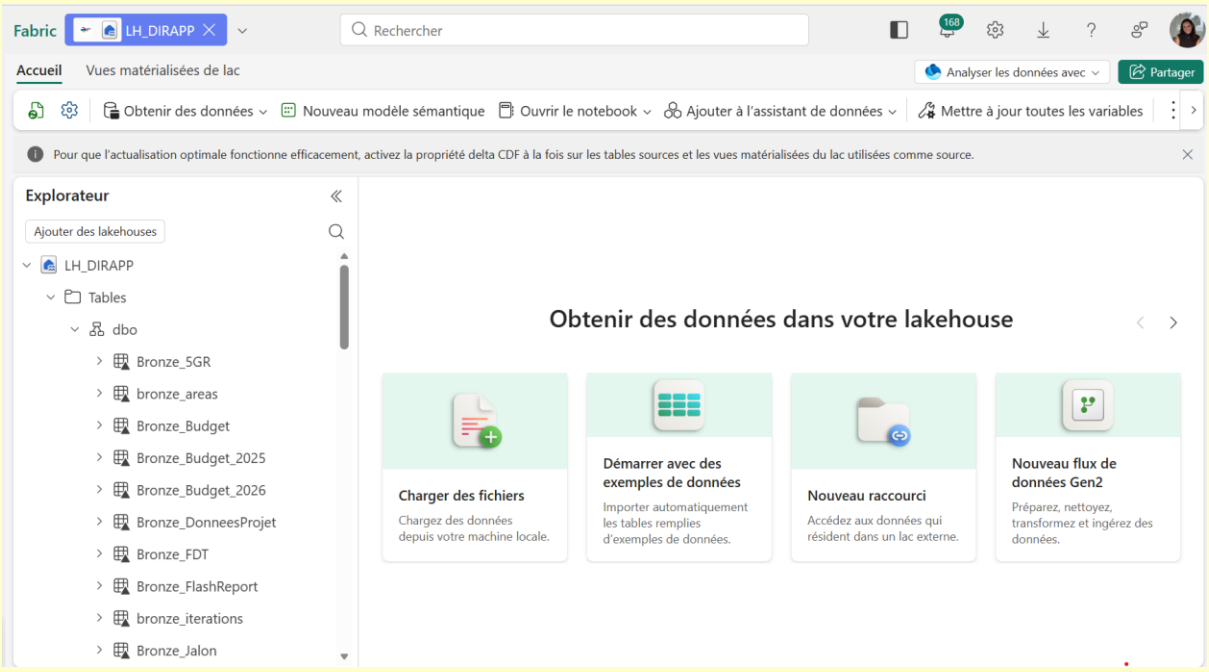
🥉 Bronze Layer — Raw data ingestion

👉 Import your dataset into the **Lakehouse**

👉 Name your tables: XX_Bronze

⚠️ **Important**
No transformation allowed here. This must be a **strict copy of your source data.**

💡 You should be able to see you table as:



🥈 Silver Layer — Data transformation

👉 Create **one Notebook per table**

👉 Clean and transform your data, make sure to:

- Load Bronze table

- Inspect schema (display())

- Fix data types

- Handle null values

- Rename columns if needed

- Save result as Silver table

👉 Name your tables: XX_Silver

⚠️ **Important**
- Do **not delete columns**
- Data must be **reusable by others**

💡 Think: "If someone needs the data → I give them Silver".

👉 Are you using an ETL or an ELT solution? Justify your answer.

🥇 **Gold Layer — Business-ready data**

💡 Ask yourself:

• Which columns are really needed for my dashboard?

• What questions must my data answer?

👉 Prepare data for the **dashboard / reporting**

👉 Name your tables Dimension_XX for dimensions and Fact_XX for facts

🧠: In a data model:

- Fact tables contain facts, quantitative data (e.g. sales, revenue)

- Dimension tables contain descriptive data (e.g. customers, products)

👉 Keep only:

- Relevant columns

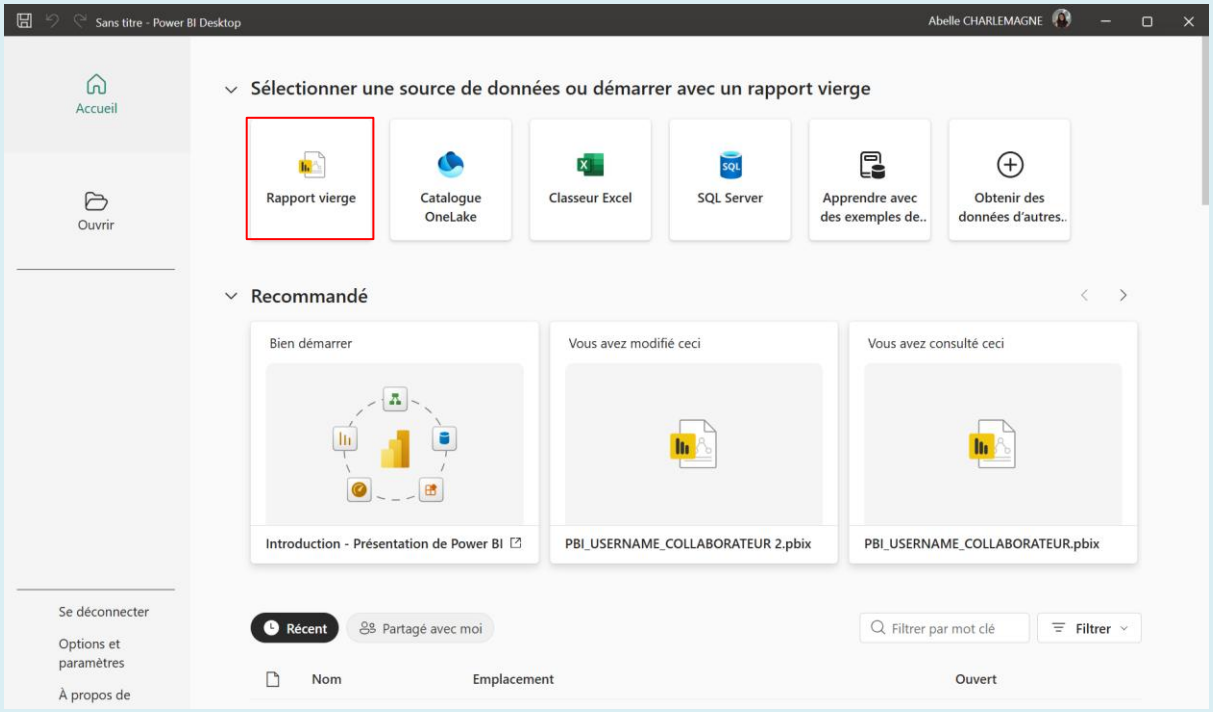
- Aggregated data if needed

⚠️ This layer should match your **business need**

💡 Think: “This is what Power BI will use”.

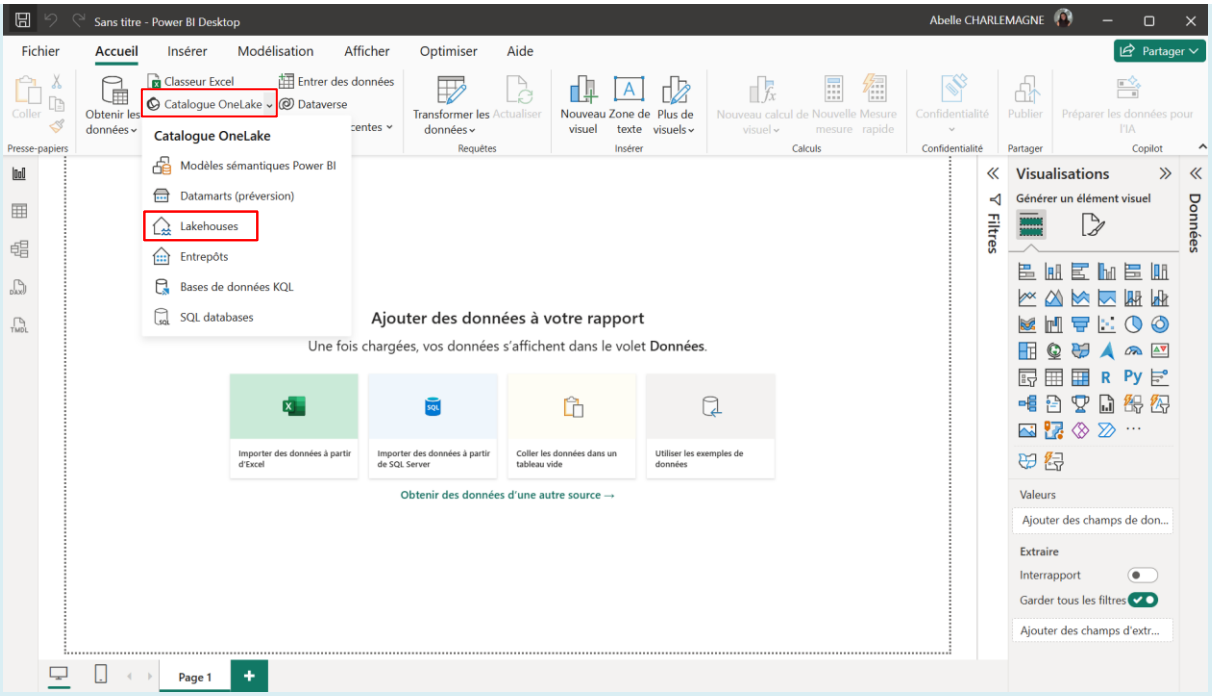
Task 4: Build your Semantic Model

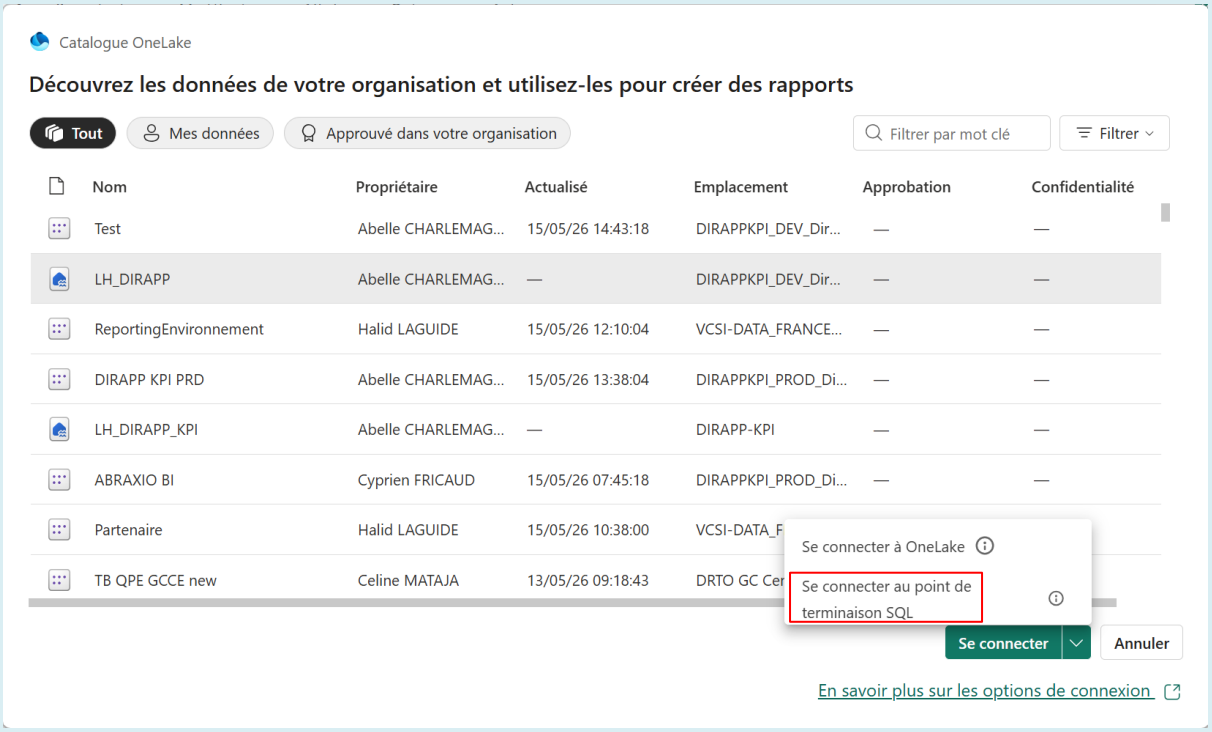
👉 Open **Power BI Desktop**, Select new report



👉 Log in with your Fabric account

👉 Connect to your Lakehouse (One Lake)

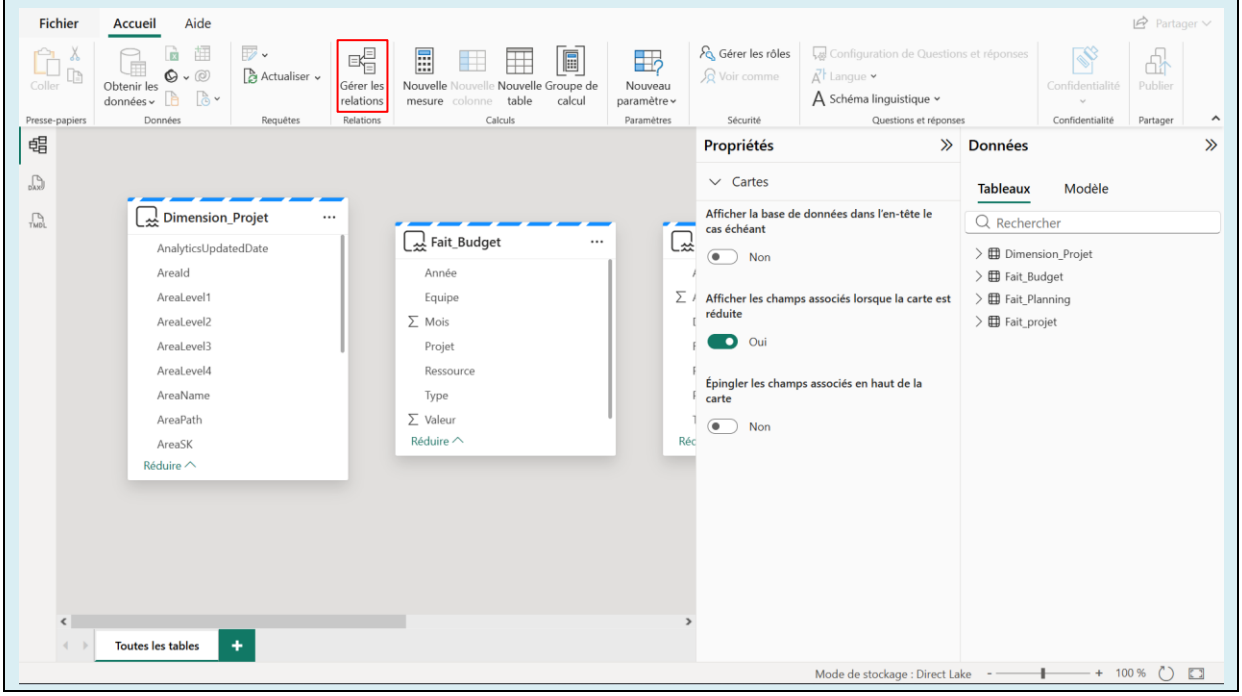




👉 Import your **Gold tables, use DirectQuery mode**

👉 Explain the difference between Import and DirectQuery mode.

👉 Create relationships between tables: Go to the **Model view** ("Manage relationships") in Power BI or by **dragging and dropping a key field** from one table to the corresponding field in another table



💡 Typically, you will link:

- A **primary key** (unique identifier)

- To a **foreign key** in another table

💡 Example:
Drag CustomerID from the *Customers* table to CustomerID in the *Orders* table

⚠️ Make sure your relationships:
- Use the correct fields
- Have the right cardinality (1-to-many in most cases)
- Have the correct cross-filter direction

👉 For each relationship in your model:

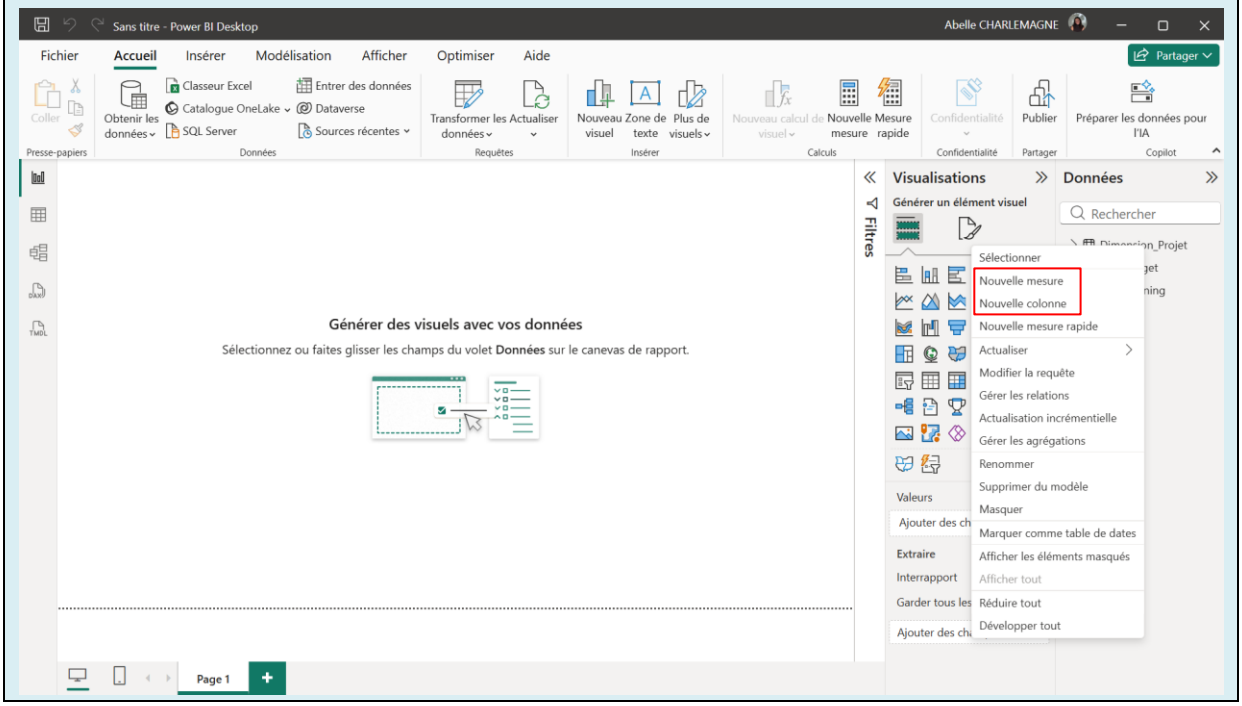
- What is the cardinality (1-to-many, many-to-many, etc.)?

- What is the filter direction?

- From which table does the filter propagate (dimension → fact or fact → dimension)?

👉 Justify your choices

👉 Add calculated column or measures if needed (e.g. Total Sales, Average Price, Growth %)



👉 Explain your semantic model, ensure your model is **clear and usable**. Describe your tables, measures, and relationships, including their cardinality.

At the end of Lab 3, you should have:

- ✅ A working Fabric Workspace
- ✅ A Bronze / Silver / Gold architecture
- ✅ Clean Silver tables
- ✅ Business-ready Gold tables
- ✅ A connected Power BI model

⭐ **Final deliverable of Lab 3**

1. Your word file
2. The link to your Fabric Workspace
3. Your Power BI file (.pbix)

💾 : Remember to upload deliverables on Moodle